LINEAR ALGEBRA

Components

$$a_i = \frac{\langle E_i|a\rangle}{\langle E_i|E_i\rangle}$$

$T_{ij} = \langle e_i|\hat{T}|e_j\rangle$ (assumes the e_i are normalized)

Inner Product

$$\langle a|b\rangle = \underline{a}^{\dagger} \cdot \underline{b}$$

$$\langle a|b\rangle = \int_{x_i}^{x_f} a^*(x)b(x)\,dx \qquad \text{for interval } [x_i, x_f]$$

Gram-Schmidt

$$|E_n\rangle = |F_n\rangle - \sum_{i=1}^{n-1} \frac{\langle E_i|F_n\rangle}{\langle E_i|E_i\rangle}\,|E_i\rangle$$

Changing Bases

$$\underline{a}^{(f)} = \underline{S} \cdot \underline{a}^{(e)}$$

$$\underline{T}^{(f)} = \underline{S} \cdot \underline{T}^{(e)} \cdot \underline{S}^{-1}$$

FOURIER ANALYSIS

Sine-cosine Series – interval $[x_0, x_0 + L]$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\frac{2\pi nx}{L} + \sum_{n=1}^{\infty} b_n \sin\frac{2\pi nx}{L}$$

$$a_0 = \frac{1}{L}\int_{x_0}^{x_0+L} f(x)\,dx$$

$$a_n = \frac{2}{L}\int_{x_0}^{x_0+L} \cos\frac{2\pi nx}{L} f(x)\,dx$$

$$b_n = \frac{2}{L}\int_{x_0}^{x_0+L} \sin\frac{2\pi nx}{L} f(x)\,dx$$

Exponential Series – interval $[x_0, x_0 + L]$

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i2\pi nx/L}$$

$$c_n = \frac{1}{L}\int_{x_0}^{x_0+L} e^{-i2\pi nx/L} f(x)\,dx$$

Fourier Transforms

$$f(x) = \int_{-\infty}^{\infty} e^{ikx} c(k)\,dk$$

$$c(k) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-ikx} f(x)\,dx$$

PARTIAL DIFFERENTIAL EQUATIONS

Separation of Variables

$$V(x,y,z) = X(x)Y(y)Z(z)$$

ODEs

$$\frac{\partial^2 X}{\partial x^2} = +k^2 X \quad \longrightarrow \quad X(x) = A\cosh kx + B\sinh kx$$
$$= A'e^{kx} + B'e^{-kx}$$

$$\frac{\partial^2 X}{\partial x^2} = -k^2 X \quad \longrightarrow \quad X(x) = A\cos kx + B\sin kx$$

OTHER EQUATIONS

Delta Functions

$$\int_{-\infty}^{\infty} f(x)\delta^{(n)}(x-a)\,dx = (-1)^n f^{(n)}(a)$$

$$\delta(kx) = \frac{1}{|k|}\delta(x) \qquad \delta\big(f(x)\big) = \sum_{\text{all roots } x_i} \frac{\delta(x-x_i)}{|f'(x_i)|}$$

Power Series

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(a)(x-a)^n$$

VECTOR CALCULUS THEOREMS

$$\int_V \vec{\nabla}\cdot\vec{F}\,dV = \oint_{\partial V} \vec{F}\cdot d\vec{A}$$

$$\int_S \left(\vec{\nabla}\times\vec{F}\right)\cdot d\vec{A} = \oint_{\partial S} \vec{F}\cdot d\vec{\ell}$$

DIFFERENTIAL OPERATORS

$$d\vec{\ell} = \hat{x}\,dx + \hat{y}\,dy + \hat{z}\,dz$$
$$= \hat{\rho}\,d\rho + \hat{\phi}\,\rho d\phi + \hat{z}\,dz$$
$$= \hat{r}\,dr + \hat{\theta}\,r d\theta + \hat{\phi}\,r\sin\theta\,d\phi$$

$$\vec{\nabla}f = \hat{x}\frac{\partial f}{\partial x} + \hat{y}\frac{\partial f}{\partial y} + \hat{z}\frac{\partial f}{\partial z}$$
$$= \hat{\rho}\frac{\partial f}{\partial \rho} + \hat{\phi}\frac{1}{\rho}\frac{\partial f}{\partial \phi} + \hat{z}\frac{\partial f}{\partial z}$$
$$= \hat{r}\frac{\partial f}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial f}{\partial \theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}$$

$$\vec{\nabla}\cdot\vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$
$$= \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho A_\rho\right) + \frac{1}{\rho}\frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$
$$= \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 A_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\,A_\theta) + \frac{1}{r\sin\theta}\frac{\partial A_\phi}{\partial \phi}$$

$$\vec{\nabla}\times\vec{A} = \hat{x}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{y}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{z}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$
$$= \hat{\rho}\left(\frac{1}{\rho}\frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z}\right) + \hat{\phi}\left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho}\right) + \hat{z}\frac{1}{\rho}\left(\frac{\partial}{\partial \rho}\left(\rho A_\phi\right) - \frac{\partial A_\rho}{\partial \phi}\right)$$
$$= \hat{r}\frac{1}{r\sin\theta}\left(\frac{\partial}{\partial \theta}\left(\sin\theta\,A_\phi\right) - \frac{\partial A_\theta}{\partial \phi}\right) + \hat{\theta}\left(\frac{1}{r\sin\theta}\frac{\partial A_r}{\partial \phi} - \frac{1}{r}\frac{\partial}{\partial r}\left(rA_\phi\right)\right) + \hat{\phi}\frac{1}{r}\left(\frac{\partial}{\partial r}\left(rA_\theta\right) - \frac{\partial A_r}{\partial \theta}\right)$$

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$
$$= \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho\frac{\partial f}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}$$
$$= \frac{1}{r}\frac{\partial^2}{\partial r^2}(rf) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial \phi^2}$$